A CONSTANT-KERNEL DIAGONAL COUNTERMODEL TO A HEAT-TRACE IMPLICATION FOR ERDŐS PROBLEM #1055

ARGUS OPEN RESEARCH PROJECT

Abstract. We exhibit an elementary all-N diagonal matrix family $H_N = \mathrm{diag}(0_K, (3/10)I_{N-K})$ showing that the bounded-kernel heat-trace inequalities in a 2025 manuscript on Erdős Problem #1055 and Cunningham chains do not imply an unbounded kernel dimension as $N \to \infty$.

1. The Countermodel

Let $K \geq 1$ be a fixed integer, and for every $N \geq K$ define the $N \times N$ positive semidefinite diagonal operator

$$H_N \;=\; \mathrm{diag}\big(\underbrace{0, \ldots, 0}_{K}, \underbrace{\tfrac{3}{10}, \ldots, \tfrac{3}{10}}_{N-K}\big).$$

Proposition 1.1. *For every fixed $K \geq 1$ and every $N \geq K$:*

(1) $\dim\ker(H_N) = K$ *is constant and bounded by* $C = K$.

(2) *The smallest positive eigenvalue of* H_N *is* $\lambda_1(H_N) = 3/10 > 0$.

(3) *For every* $t > 0$, *the heat trace satisfies*

$$(N - C)e^{-3t/10} \;\leq\; \mathrm{Tr}(e^{-tH_N}) \;=\; K + (N - K)e^{-3t/10} \;\leq\; C + (N - C)e^{-3t/10}.$$

The upper and lower bounds in (3) differ by $C = K > 0$; *equating them in the limit* $N \to \infty$ *without dividing by* N *is invalid, and the inequalities are satisfied by a family with constant kernel dimension* K.

Date: August 29, 2026.

2020 *Mathematics Subject Classification.* Primary 11A41; Secondary 15A18.